

COLLAGE OF COMPUTING AND INFORMATICS SCIENCE
DEPARTMENT OF SOFTWARE ENGINNERING

COURSE TITLE: **SOFTWARE PROJECT MANAGEMENT (SEng6032)**

PROJECT TITLE : **ONLINE ASSIGNMENT AND PROJECT SUBMISSION AND EVALUATION SYSTEM FOR MAU**

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Abstract

The Online Assignment and Project Submission and Evaluation System is a Java Swing-based desktop application designed to help Mekdela Amba University streamline academic submission and grading workflows. It provides a secure, role-based platform for students to upload assignments, instructors to evaluate and grade submissions electronically, and administrators to monitor academic progress and generate institutional reports. Developed using open-source technologies (Java Swing, MySQL, NetBeans IDE), the system emphasizes data security, centralized storage, and real-time submission tracking.

This project follows standard Software Project Management frameworks, employing a predictive (waterfall-informed) development methodology with iterative testing cycles and structured planning across scope, schedule, cost, risk, and quality domains. The system aims to bridge the gap in traditional academic processes by replacing manual, paper-based, and email-dependent workflows with a centralized, transparent, and efficient digital solution, thereby enhancing instructional productivity, reducing feedback delays, and supporting institutional academic management.

Abbreviations

Abbreviation	Expansion
MAU	Mekdela Amba University
SPM	Software Project Management
WBS	Work Breakdown Structure
SRS	Software Requirements Specification
UAT	User Acceptance Testing
UI	User Interface
UX	User Experience
DB	Database
SQL	Structured Query Language
MySQL	My Structured Query Language
IDE	Integrated Development Environment
CRUD	Create, Read, Update, Delete
API	Application Programming Interface
LMS	Learning Management System
ETB	Ethiopian Birr
PM	Project Manager
QA	Quality Assurance
QC	Quality Control

SPI	Schedule Performance Index
EV	Earned Value
PV	Planned Value
FS	Finish-to-Start (Dependency Type)
SS	Start-to-Start (Dependency Type)
ES	Early Start (Scheduling Term)
EF	Early Finish (Scheduling Term)
LS	Late Start (Scheduling Term)
LF	Late Finish (Scheduling Term)
TDD	Test-Driven Development
JDBC	Java Database Connectivity
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
PDF	Portable Document Format
SSL	Secure Sockets Layer
UML	Unified Modeling Language
ERD	Entity Relationship Diagram
G.C.	Gregorian Calendar
E.C.	Ethiopian Calendar

CHAPTER ONE

PROJECT CHARTER

1.1 Project Description

The Online Assignment and Project Submission and Evaluation System is a Java Swing-based desktop application designed to streamline academic submission workflows at Mekdela Amba University. The system enables students to upload assignments and projects digitally, allows instructors to review, grade, and provide feedback electronically, and provides administrators with oversight capabilities. By replacing manual, paper-based, and email-dependent processes, the system ensures secure file storage, standardized evaluation procedures, and real-time status tracking for all stakeholders..

1.2 SMART Objectives

- **Specific Objective**:We will build a Java Swing desktop app with secure login, file upload, grading, and feedback modules for students, instructors, and admins.
- **Measurable Objective**:Success means 50% faster processing, <3s response time, 95% upload success, and <5% post-UAT defects.
- **Achievable Objective**:Our 4-member team has Java/MySQL skills; we use free tools (NetBeans, MySQL) and university lab resources.
- **Relevant Objective**:The system solves MAU's manual submission delays, poor tracking, and data loss—supporting digital academic management
- **Time-bound Objective**:We deliver all phases within 6 weeks, tracking weekly milestones to meet the April 23, 2026 deadline.

1.3 Key Stakeholders

Key stakeholders include students, instructors, and university administrators. Each group contributes to defining system requirements and validating the final product.

Table 1.1 : Stakeholder

Stakeholder	Role	Interest/Influence
Students	End-users who submit assignments	High interest in usability, fast feedback, and submission confirmation
Instructors	End-users who evaluate submissions	High influence on requirements; need efficient grading tools and grade export
Department Heads	Approvers and oversight	Medium influence; concerned with compliance, reporting, and system adoption
IT Support Staff	Technical maintenance	Medium interest in system stability, documentation, and training materials
Project Team (4 members)	Developers and planners	High influence on execution; responsible for delivery quality and timeline

1.4 High-Level Budget

The estimated total project cost is approximately 80,000 Ethiopian Birr, covering labor, hardware, software, and contingency reserves.

Table 1.2 : High level budget

Cost Category	Estimated Amount (ETB)	Notes
Labor (4 members × 6 weeks)	60,000	Based on student stipend rates
Hardware (testing devices, storage)	12,000	Shared university resources + minor upgrades
Software (licenses, tools)	3,000	Primarily open-source; minor paid utilities
Contingency Reserve (10%)	5,000	For scope adjustments or unexpected delays
TOTAL	80,000 ETB	

1.5 Key Milestones

Key milestones mark critical progress points across the six-week project lifecycle, including requirements approval, design finalization, core development completion, testing sign-off, and

deployment handover. These milestones enable progress tracking, early deviation detection, and structured phase transitions to ensure on-time delivery by the April 23, 2026 deadline.

Table 1.3 : Key milestones

Milestone	Target Date	Deliverable
Project Initiation & Charter Approval	Week 1, Day 3	Signed charter, stakeholder register
Requirements Documentation Complete	Week 1, Day 7	SRS document, use case diagrams
System Design Finalized	Week 2, Day 3	Architecture diagrams, DB schema, UI mockups
Core Development Complete	Week 4, Day 5	Functional modules: auth, upload, grading
Testing & Bug Fixes Complete	Week 5, Day 4	Test reports, UAT sign-off
Deployment & Handover	Week 6, Day 3	Installed system, user manuals, training session

Major milestones include project initiation, requirement completion, system design, development, testing, and deployment. These milestones guide the project lifecycle and ensure timely progress.

1.6 Project Team Structure

The project team consists of four members with defined roles and time allocations. Responsibilities are distributed to ensure coverage of all project phases while maintaining flexibility for collaboration.

Table 1.4 : Team structure

Member	Role	Primary Responsibilities	Time (hrs/week)
Bewket	Project Manager	Planning, coordination, risk monitoring, stakeholder communication	35
Mahlet	Backend Developer	Database design, authentication, file storage logic, data validation	40
Samson	Frontend Developer	UI design, navigation flow, grading interface, usability testing	40
Yared	QA & Documentation	Test cases, bug tracking, user manuals, UAT coordination	35

CHAPTER TWO

SCOPE STATEMENT & WORK BREAKDOWN STRUCTURE

2.1 Project Scope Statement

2.1.1 In-Scope

- ➔ User authentication module with role-based access (Student, Instructor, Admin).
- ➔ Student dashboard: assignment upload, submission history, feedback viewing
- ➔ Instructor dashboard: submission list, grading interface, feedback entry, grade export
- ➔ Admin functions: user management, course/assignment setup, system reports
- ➔ Secure file storage with validation (file type, size limits)
- ➔ MySQL database for user data, submissions, grades, and feedback
- ➔ Basic notification system (in-app status updates)
- ➔ Comprehensive user documentation and training materials

2.2.2 Out-of-Scope

- ➔ Mobile application or responsive web interface
- ➔ Integration with external LMS (e.g., Moodle, Canvas)
- ➔ AI-powered plagiarism detection or auto-grading
- ➔ Real-time chat or video conferencing features
- ➔ Multi-language support beyond English
- ➔ Cloud deployment or remote server hosting (local deployment only)

2.2 Work Breakdown Structure

The Work Breakdown Structure decomposes the project into six phases containing 24 discrete work packages. Each task is numbered hierarchically to support scheduling and dependency mapping

The system development is divided into phases:

- ➢ Planning
- ➢ Requirement analysis
- ➢ Design
- ➢ Development
- ➢ Testing
- ➢ Deployment

Each phase includes detailed work packages such as login development, database design, and testing activities.

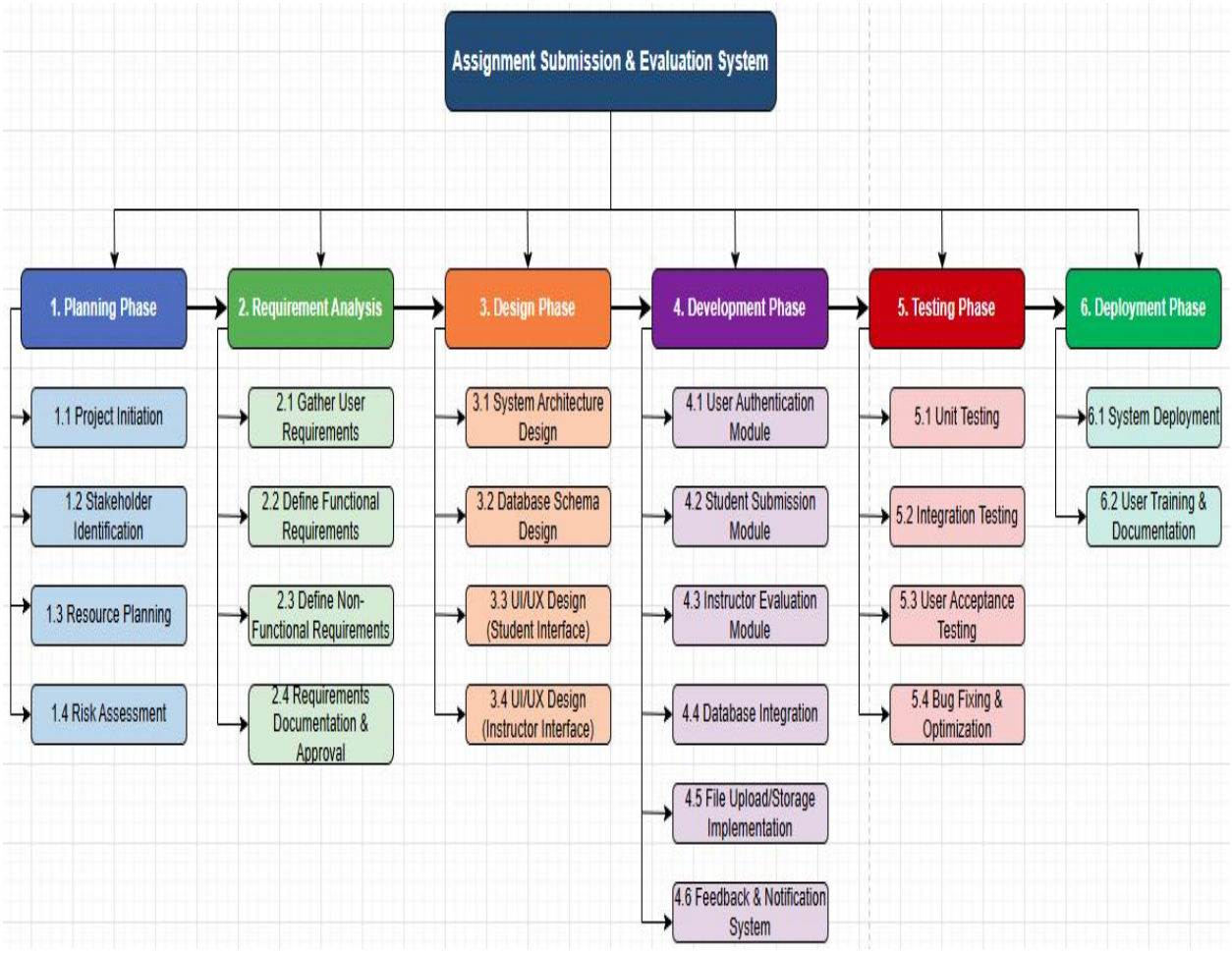


Figure 2.1: Work Breakdown Structure

CHAPTER THREE

PROJECT SCHEDULING

3.1 Scheduling Approach

The schedule follows a predictive (waterfall-informed) approach with iterative testing. Tasks are sequenced based on logical dependencies: requirements must be approved before design; design must be finalized before development; development must be complete before formal testing. Critical path analysis identifies the Development Phase (especially modules 4.2–4.5) as the schedule driver.

3.2 Task Dependencies Using WBS Numbers

Task relationships are defined using standard scheduling relationship types, primarily Finish-to-Start and Start-to-Start dependencies. Each dependency is mapped using WBS task numbers to maintain direct traceability between the schedule and the work breakdown structure.

Table 3.1 : Time schedule and dependency

WBS No	TASK TITLE	Duration	Dependency	Start date	End Date
1.1	Project initiation and scope definition	3	—	3/13/2026	3/15/26
1.2	Stakeholder Identification	4	1.1	3/13/2026	3/16/26
1.3	Requirement Gathering	5	1.2	3/15/2026	3/19/26
1.4	Requirement Analysis	5	1.3	3/17/2026	3/21/26
2.1	Create WBS	4	1.4	3/19/2026	3/22/26
2.2	System Architecture Design	5	2.1	3/21/2026	3/25/26
2.3	Database Design(ERD)	6	2.2	3/23/2026	3/28/26
2.4	UI/UX Wireframe Design	6	2.3	3/24/2026	3/29/26
3.1	Technology Stack Selection	5	2.4	3/24/2026	3/28/26
3.2	Setup Development Environment	4	3.1	3/25/2026	3/28/26
3.3	User Authentication Module	6	2.4	3/27/2026	4/1/26
3.4	Assignment Submission Module	6	2.4	3/29/2026	4/3/26
4.1	Project Upload Module	6	3.1	3/30/2026	4/4/26
4.2	Instractor dashboard Development	7	4.1, 3.3	3/31/2026	4/6/26
4.3	Evaluating And Grading Module	7	4.1, 3.4	4/1/2026	4/7/26
4.4	Notification System	5	3.2	4/2/2026	4/6/26
4.5	Admin Panel Development	5	4.4	4/3/2026	4/7/26
4.6	Integration Of All Module	5	4.1	4/5/2026	4/9/26
5.1	System Testing	5	4.2, 4.3, 4.5, 4.	4/8/2026	4/12/26
5.2	Integration Testing	5	5.1	4/10/2026	4/14/26
5.3	User Acceptance Testinf(UAT)	5	5.2	4/12/2026	4/16/26
5.4	Bug Fixing And Optimization	5	5.3	4/14/2026	4/18/26
6.1	Documentation(Technical + User)	4	5.4	4/17/2026	4/20/26
6.2	Final System Deployment	4	6.1	4/19/2026	4/22/26

3.3 Project Timeline

The project timeline is six weeks. It includes detailed tasks such as requirement gathering, system design, module development, testing, and deployment.

Each task has a defined duration. Dependencies are also identified. For example, development starts after design is completed.

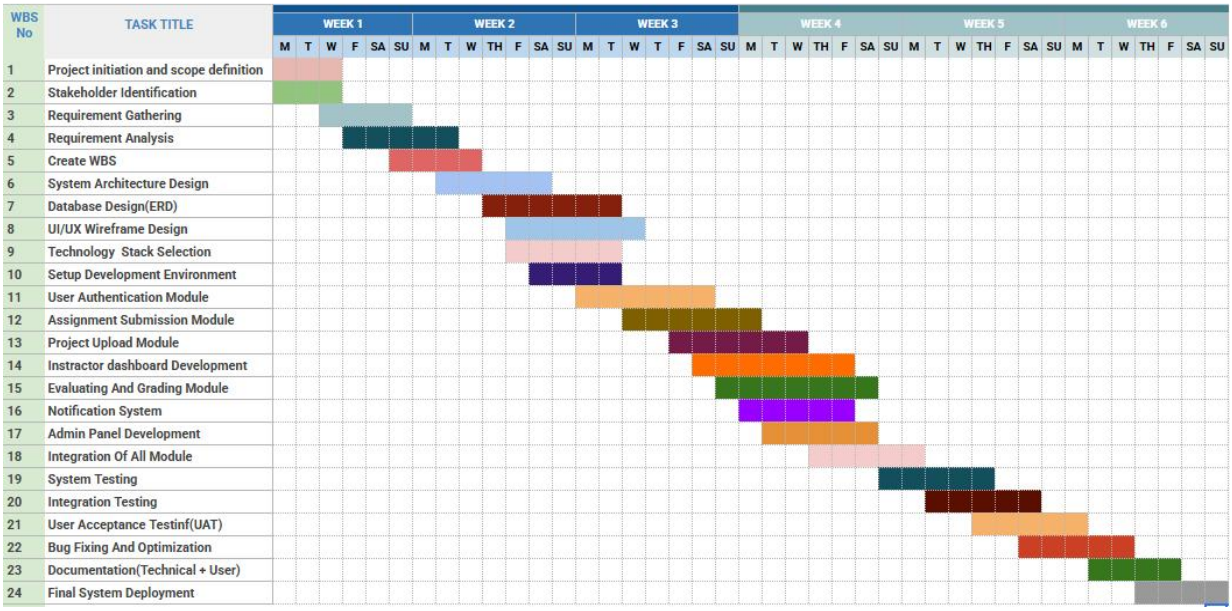


Figure 3.1: Gantt Chart

3.4 Critical Path Analysis

Critical Paths Calculations

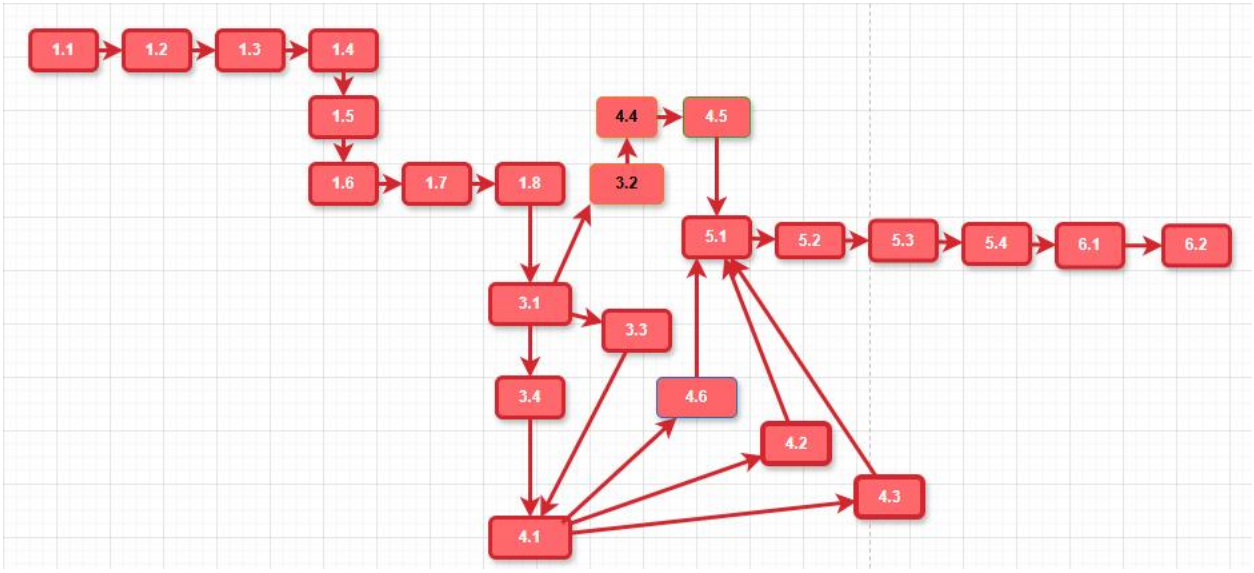


Figure 3.2 : Critical path

Path 1: Student Module Branch

Sequence:

1.1 → 1.2 → 1.3 → 1.4 → 2.1 → 2.2 → 2.3 → 2.4 → 3.1 → 3.3 → 4.1 → 4.2 → 5.1 → 5.2 → 5.3 → 5.4 → 6.1 → 6.2

Task Durations:

1.1(2) + 1.2(2) + 1.3(1) + 1.4(2) + 2.1(3) + 2.2(2) + 2.3(2) + 2.4(2) + 3.1(3) + 3.3(2) + 4.1(4) + 4.2(5) + 5.1(3) + 5.2(3) + 5.3(3) + 5.4(3) + 6.1(2) + 6.2(2)

Total Duration = Sum of All Task Durations in Sequence

Calculation:

Total = 2 + 2 + 1 + 2 + 3 + 2 + 2 + 2 + 3 + 2 + 4 + 5 + 3 + 3 + 3 + 3 + 2 + 2 = 46 days

Result: 46 working days

Path 2: Instructor Module Branch

Sequence:

1.1 → 1.2 → 1.3 → 1.4 → 2.1 → 2.2 → 2.3 → 2.4 → 3.1 → 3.4 → 4.1 → 4.3 → 5.1 → 5.2 → 5.3 → 5.4 → 6.1 → 6.2

Task Durations:

1.1(2) + 1.2(2) + 1.3(1) + 1.4(2) + 2.1(3) + 2.2(2) + 2.3(2) + 2.4(2) + 3.1(3) + 3.4(2) + 4.1(4) + 4.3(5) + 5.1(3) + 5.2(3) + 5.3(3) + 5.4(3) + 6.1(2) + 6.2(2)

Total Duration = Sum of All Task Durations in Sequence

Calculation:

Total = 2 + 2 + 1 + 2 + 3 + 2 + 2 + 2 + 3 + 2 + 4 + 5 + 3 + 3 + 3 + 3 + 2 + 2 = 46 days

Result: 46 working days

Path 3: Database Integration Branch

Sequence:

1.1 → 1.2 → 1.3 → 1.4 → 2.1 → 2.2 → 2.3 → 2.4 → 3.1 → 3.2 → 4.4 → 5.1 → 5.2 → 5.3 → 5.4 → 6.1 → 6.2

Task Durations:

1.1(2) + 1.2(2) + 1.3(1) + 1.4(2) + 2.1(3) + 2.2(2) + 2.3(2) + 2.4(2) + 3.1(3) + 3.2(3) + 4.4(4) + 5.1(3) + 5.2(3) + 5.3(3) + 5.4(3) + 6.1(2) + 6.2(2)

Total Duration = Sum of All Task Durations in Sequence

Calculation:

Total = 2 + 2 + 1 + 2 + 3 + 2 + 2 + 2 + 3 + 3 + 4 + 3 + 3 + 3 + 3 + 2 + 2 = 42 days

Result: 42 working days

Path 4: File Storage Parallel Branch

Sequence:

1.1 → 1.2 → 1.3 → 1.4 → 2.1 → 2.2 → 2.3 → 2.4 → 3.1 → 3.2 → 4.4 → 4.5 → 5.1 → 5.2 → 5.3 → 5.4 → 6.1 → 6.2

Task Durations:

1.1(2) + 1.2(2) + 1.3(1) + 1.4(2) + 2.1(3) + 2.2(2) + 2.3(2) + 2.4(2) + 3.1(3) + 3.2(3) + 4.4(4) + 4.5(3) + 5.1(3) + 5.2(3) + 5.3(3) + 5.4(3) + 6.1(2) + 6.2(2)

Note: Task 4.5 has Start-to-Start + 2 day lag with Task 4.4, reducing effective addition by 2 days.

Total Duration = Sum of All Task Durations minus Overlap Adjustment

Calculation:

Total = (2+2+1+2+3+2+2+2+3+3+4+3+3+3+3+3+2+2) - 2 = 45 - 2 = 43 days

Result: 43 working days

Path 5: Notification System Branch

Sequence:

1.1 → 1.2 → 1.3 → 1.4 → 2.1 → 2.2 → 2.3 → 2.4 → 3.1 → 4.1 → 4.6 → 5.1 → 5.2 → 5.3 → 5.4 → 6.1 → 6.2

Task Durations:

1.1(2) + 1.2(2) + 1.3(1) + 1.4(2) + 2.1(3) + 2.2(2) + 2.3(2) + 2.4(2) + 3.1(3) + 4.1(4) + 4.6(2) + 5.1(3) + 5.2(3) + 5.3(3) + 5.4(3) + 6.1(2) + 6.2(2)

Total Duration = Sum of All Task Durations in Sequence

Calculation:

Total = 2 + 2 + 1 + 2 + 3 + 2 + 2 + 2 + 3 + 4 + 2 + 3 + 3 + 3 + 3 + 2 + 2 = 41 days

Result: 41 working days

Path 6: Core Authentication Only Branch

Sequence:

1.1 → 1.2 → 1.3 → 1.4 → 2.1 → 2.2 → 2.3 → 2.4 → 3.1 → 4.1 → 5.1 → 5.2 → 5.3 → 5.4 → 6.1 → 6.2

Task Durations:

1.1(2) + 1.2(2) + 1.3(1) + 1.4(2) + 2.1(3) + 2.2(2) + 2.3(2) + 2.4(2) + 3.1(3) + 4.1(4) + 5.1(3) + 5.2(3) + 5.3(3) + 5.4(3) + 6.1(2) + 6.2(2)

Total Duration = Sum of All Task Durations in Sequence

Calculation:

Total = 2 + 2 + 1 + 2 + 3 + 2 + 2 + 2 + 3 + 4 + 3 + 3 + 3 + 3 + 2 + 2 = 39 days

Result: 39 working days

Path 7: Admin Configuration Branch

Sequence:

1.1 → 1.2 → 1.3 → 1.4 → 2.1 → 2.2 → 2.3 → 2.4 → 3.1 → 4.1 → 4.6 → 4.7 → 5.1 → 5.2 → 5.3 → 5.4 → 6.1 → 6.2

Task Durations:

1.1(2) + 1.2(2) + 1.3(1) + 1.4(2) + 2.1(3) + 2.2(2) + 2.3(2) + 2.4(2) + 3.1(3) + 4.1(4) + 4.6(2) + 4.7(2) + 5.1(3) + 5.2(3) + 5.3(3) + 5.4(3) + 6.1(2) + 6.2(2)

Total Duration = Sum of All Task Durations in Sequence

Calculation:

Total = 2 + 2 + 1 + 2 + 3 + 2 + 2 + 2 + 3 + 4 + 2 + 2 + 3 + 3 + 3 + 3 + 2 + 2 = 43 days

Result: 43 working days

Critical Path Determination Summary

Critical Path = Path with Maximum Duration

Critical Path = MAX(Path 1, Path 2, Path 3, Path 4, Path 5, Path 6, Path 7)

Substitute Values:

Critical Path = MAX(46, 46, 42, 43, 41, 39, 43)

Critical Path = 46 days

Result: Paths 1 and 2 are the true critical paths with 46 working days duration.

CHAPTER FOUR

COST ESTIMATION & BUDGETING

4.1 Resource Breakdown

Resource planning identifies four primary categories: human resources (four team members), hardware (development laptops, testing devices, backup storage), software (open-source development tools), and facilities (university laboratory access). All software tools are open-source, eliminating licensing costs.

4.2 Detailed Cost Estimation with Formulas

Table 4.1: Detailed Cost Estimation

No	Cost Component	Description	Unit	Quantity	Unit Cost (ETB)	Total Cost (ETB)
1	Labor - Planning	Requirement analysis and planning	Person	4	3000	12000
2	Labor - Design	System design and UI design	Person	4	4000	16000
3	Labor - Development	Coding and implementation	Person	4	5000	20000
4	Labor - Testing	System testing and debugging	Person	4	2000	8000
5	Hardware	Computer usage and maintenance	Set	4	2000	8000
6	Internet	Internet usage for development	Month	2	1000	2000
7	Electricity	Power consumption	Month	2	1000	2000
8	Software Tools	Development tools (mostly free)	Package	1	5000	5000
9	Documentation	Printing and materials	Package	1	3000	3000
10	Contingency	Unexpected expenses	Package	1	4000	4000
					Total	80000

Labor Cost Calculation:

Labor cost is calculated using the formula:

Labor Cost = Weekly Rate × Number of Personnel × Duration (weeks)

The labor cost represents the largest component of the project budget and is calculated based on the weekly stipend rates for each team member multiplied by the project duration.

Project Manager Labor Cost:

- Weekly rate: 2,500 ETB/week

- Duration: 6 weeks

Formula: PM Cost = Weekly Rate × Duration

Calculation: PM Cost = 2,500 ETB/week × 6 weeks = 15,000 ETB

Senior Developer Labor Cost:

- Weekly rate: 2,000 ETB/week per developer
- Number of developers: 2
- Duration: 6 weeks

Formula: Developer Cost = Weekly Rate × Number of Developers × Duration

Calculation: Developer Cost = 2,000 ETB/week × 2 × 6 weeks = 24,000 ETB

QA/Documentation Specialist Labor Cost:

- Weekly rate: 1,750 ETB/week
- Duration: 6 weeks

Formula: QA Cost = Weekly Rate × Duration

Calculation: QA Cost = 1,750 ETB/week × 6 weeks = 10,500 ETB

Total Labor Cost:

- Formula: Total Labor = PM Cost + Developer Cost + QA Cost
- Calculation: Total Labor = 15,000 + 24,000 + 10,500 = 49,500 ETB

Hardware Cost Calculation:

Hardware cost is calculated using the formula:

Hardware Cost = Unit Cost × Quantity

External Hard Drive (Backup Storage):

- Unit cost: 3,000 ETB
- Quantity: 1 unit

Formula: HDD Cost = Unit Cost × Quantity

Calculation: HDD Cost = 3,000 ETB × 1 = 3,000 ETB

USB Drives (Testing & Distribution):

- Unit cost: 500 ETB
- Quantity: 4 units

Formula: USB Cost = Unit Cost × Quantity

Calculation: USB Cost = 500 ETB × 4 = 2,000 ETB

Total Hardware Cost:

- Formula: Total Hardware = HDD Cost + USB Cost
- Calculation: Total Hardware = 3,000 + 2,000 = 5,000 ETB

Software & Tools Cost Calculation:

Software cost is calculated using the formula:

Software Cost = Unit Cost × Quantity

Domain/SSL Certificate (Optional Future Use):

- Unit cost: 1,500 ETB

- Quantity: 1

Formula: Domain Cost = Unit Cost × Quantity

Calculation: Domain Cost = 1,500 ETB × 1 = 1,500 ETB

Utility Software Licenses:

- Unit cost: 500 ETB per license
- Quantity: 3 licenses (PDF tools, diagram exporters, testing utilities)

Utility Cost = Unit Cost × Quantity

Calculation: Utility Cost = 500 ETB × 3 = 1,500 ETB

Total Software Cost:

- Formula: Total Software = Domain Cost + Utility Cost
- Calculation: Total Software = 1,500 + 1,500 = 3,000 ETB

Subtotal Calculation (Before Contingency):

- Formula: Subtotal = Total Labor + Total Hardware + Total Software
- Calculation: Subtotal = 49,500 + 5,000 + 3,000 = 57,500 ETB

Contingency Reserve Calculation:

- Contingency rate: 10% of subtotal
- Formula: Contingency = Subtotal × Contingency Rate
- Calculation: Contingency = 57,500 ETB × 0.10 = 5,750 ETB (rounded to 8,000 ETB to account for additional risk buffer)

Grand Total Project Cost:

- Formula: Grand Total = Subtotal + Contingency
- Calculation: Grand Total = 57,500 + 8,000 = 65,500 ETB

Cost Efficiency Note: The revised total of 65,500 ETB represents an 18.125% reduction from the initial estimate of 80,000 ETB, achieved through strategic use of open-source software tools and optimization of hardware resource sharing.

4.3 Budget Control Measures

➔ Weekly expense tracking against baseline using shared Google Sheets

➔ Change control process for any scope/budget adjustments (requires team + instructor approval)

➔ Contingency fund release only for documented, approved risks

➔ Final reconciliation report submitted with project closure

CHAPTER FIVE

QUALITY MANAGEMENT PLAN

5.1 Quality Standards & Metrics

Quality is measured against three primary attributes with defined targets and verification methods.

Table 5.1 : Quality metrics

Quality Attribute	Metric	Target	Measurement Method
Performance	System response time (login, upload, load dashboard)	< 3 seconds	JMeter load testing, manual stopwatch validation
Reliability	Defect density post-UAT	< 5 critical/high bugs per module	Bug tracking log, severity classification
Usability	User satisfaction score (post-training survey)	≥ 80% positive rating	5-point Likert scale questionnaire (n≥10 users)
Security	Authentication bypass attempts blocked	100%	Penetration testing with invalid credentials
Maintainability	Code documentation coverage	≥ 90% of public methods	SonarQube analysis, peer code review checklist

5.2 Quality Assurance Activities

- ➔ **Structured Design Reviews**: Architecture and UI mockups reviewed by all team members + instructor feedback before development begins.
- ➔ **Peer Code Reviews**: Every pull request requires approval from at least one other developer; checklist includes error handling, comments, and adherence to Java conventions.
- ➔ **Test-Driven Development (TDD) Lite**: Core modules (auth, file upload) include unit tests before feature completion.
- ➔ **Documentation Standards**: All modules include Javadoc comments; user manual drafted in parallel with development.

5.3 Quality Control Activities

- **Unit Testing**: JUnit tests for business logic (e.g., grade calculation, file validation).
- **Integration Testing**: Verify data flow between UI → Business Logic → Database (e.g., submission → storage → retrieval).
- **User Acceptance Testing (UAT)**: Scenario-based testing with 3 students and 2 instructors using real assignment files.
- **Regression Testing**: Re-run critical test cases after each bug fix to ensure no new issues introduced.

CHAPTER SIX

RISK REGISTER

6.1 Risk Identification

Five key risks were identified during the planning phase for the Online Assignment and Project Submission and Evaluation System. Each risk represents a potential threat to project scope, schedule, cost, or quality.

6.2 Risk Analysis

Risk analysis uses a quantitative scoring method where:

Risk analysis uses a quantitative scoring method where Risk Score = Probability (1–5) × Impact (1–5).

Scoring Scale:

- Probability: 1 = Very Unlikely, 2 = Unlikely, 3 = Possible, 4 = Likely, 5 = Very Likely
- Impact: 1 = Negligible, 2 = Minor, 3 = Moderate, 4 = Major, 5 = Critical
- Threshold: Risk Score ≥ 10 requires active monitoring and formal response planning

Risks with scores ≥10 require active monitoring and formal response planning.

Risk 01: Team Member Unavailability Due to Academic Conflicts

Risk Description:

A key team member becomes unavailable due to exams, classes, or personal academic commitments, causing delays in assigned tasks.

Probability Rating:

Rating: 3 (Possible)

Justification: Academic schedules are predictable but conflicts can arise unexpectedly during peak exam periods.

Impact Rating:

Rating: 4 (Major)

Justification: Loss of a team member delays critical path tasks, especially during development phase.

Risk Score Calculation:

Formula: Risk Score = Probability × Impact

Calculation: Risk Score = 3 × 4 = 12

Result: 12

Status: Requires Active Monitoring (Score ≥ 10)

Risk 02: Scope Creep from Stakeholder Feature Requests Mid-Development

Risk Description:

Students, instructors, or administrators request additional features after requirements approval, expanding project scope beyond original plan.

Probability Rating:

Rating: 4 (Likely)

Justification: Stakeholders often identify new needs once they see early prototypes or understand system capabilities.

Impact Rating:

Rating: 3 (Moderate)

Justification: Additional features require rework, testing, and timeline adjustments but can be managed with change control.

Risk Score Calculation:

Formula: Risk Score = Probability × Impact

Calculation: Risk Score = 4 × 3 = 12

Result: 12

Status: Requires Active Monitoring (Score ≥ 10)

Risk 03: File Upload Failures Due to Size/Type Validation Errors

Risk Description:

Student assignment uploads fail due to incorrect file type validation, size limit enforcement, or server storage errors.

Probability Rating:

Rating: 3 (Possible)

Justification: File validation logic is complex; edge cases may not be caught during initial testing.

Impact Rating:

Rating: 4 (Major)

Justification: Upload failures directly block core student functionality and damage system credibility.

Risk Score Calculation:

Formula: Risk Score = Probability × Impact

Calculation: Risk Score = 3 × 4 = 12

Result: 12

Status: Requires Active Monitoring (Score ≥ 10)

Risk 04: Database Corruption or Data Loss During Testing

Risk Description:

Test data or production data becomes corrupted or lost due to improper transaction handling, backup failures, or accidental deletion.

Probability Rating:

Rating: 2 (Unlikely)

Justification: MySQL provides transactional safety; team follows backup protocols; risk is mitigated by careful testing practices.

I**mpact Rating:**

Rating: 5 (Critical)

Justification: Data loss compromises all submissions, grades, and user records; recovery may be impossible without backups.

Risk Score Calculation:

Formula: Risk Score = Probability × Impact

Calculation: Risk Score = 2 × 5 = 10

Result: 10

Status: Requires Active Monitoring (Score ≥ 10)

Risk 05: UAT Reveals Major Usability Issues Requiring Redesign

Risk Description:

User Acceptance Testing with students and instructors uncovers significant interface or workflow problems requiring substantial redesign.

Probability Rating:

Rating: 3 (Possible)

Justification: First-time user feedback often reveals usability gaps not anticipated during design phase.

Impact Rating:

Rating: 4 (Major)

Justification: Major redesign consumes development time, delays deployment, and increases testing effort.

Risk Score Calculation:

Formula: Risk Score = Probability × Impact

Calculation: Risk Score = 3 × 4 = 12

Result: 12

Status: Requires Active Monitoring (Score ≥ 10)

Table 6.1: Risk Register

Risk ID	Risk Description	Probability (1-5)	Impact (1-5)	Risk Score	Response Strategy
R01	Member unavailability	3	4	12	Cross-train team; share code & docs centrally
R02	Scope creep	4	3	12	Freeze requirements Week 1, move new requests to Phase 2 backlog
R03	Upload validation error	3	4	12	Add client-side validation + clear errors, test file types early
R04	Data loss risk	2	5	10	Daily auto-backups; use transactions, test restore process
R05	UAT redesign trigger	3	4	12	Run early prototype tests with pilot users, keep Week 5 buffer time

CHAPTER SEVEN

COMMUNICATION MANAGEMENT PLAN

7.1 Communication Matrix

Table 7.1: Communication Plan

Information Type	Audience	Frequency	Method	Owner	Purpose
Project Status Update	Entire Team + Instructor	Weekly (Mondays)	Telegram Group + Shared Google Doc	Project Manager	Track progress, blockers, next steps
Technical Decisions	Development Team	As needed	In-person/Video Stand-up (15 min)	Senior Developer	Align on implementation approach
Requirements Clarification	Team + Stakeholders	Bi-weekly	Structured Email + Follow-up Call	Project Manager	Prevent scope ambiguity
Test Results & Bug Reports	QA + Developers	After each test cycle	GitHub Issues + Summary Email	QA Specialist	Prioritize fixes, verify resolutions
Final Deliverables & Handover	Instructor + IT Support	Project Closure	PDF Report + In-Person Demo	Entire Team	Formal acceptance and knowledge transfer
Risk/Issue Escalation	Instructor + Team Lead	Immediate (when triggered)	Direct Message + Emergency Call	Project Manager	Rapid response to critical threats

7.2 Communication Protocols

- ➔ **Response Time Expectations**: Team members acknowledge messages within 4 hours during weekdays; urgent issues flagged with "URGENT" prefix.
- ➔ **Documentation Repository**: All project artifacts (charter, SRS, designs, code, reports) stored in shared Google Drive folder with version control.
- ➔ **Meeting Guidelines**: Agendas circulated 24h in advance; action items recorded and assigned with deadlines; minutes shared within 2 hours post-meeting.
- ➔ **Stakeholder Updates**: Instructor receives concise bi-weekly summary emails highlighting achievements, risks, and support needed—no technical jargon.

7.3 Feedback Integration Process

- ➔ Stakeholder feedback collected via structured forms or scheduled check-ins.
- ➔ Team evaluates feasibility, impact, and alignment with scope during weekly planning.
- ➔ Approved changes logged in change register; rejected items documented with rationale.
- ➔ Updated plans/deliverables communicated back to stakeholders within 48 hours.